$$\widehat{\psi}_w - \Psi_N = S_{\text{des}} + T_1 + T_2, \qquad \Psi_N - \Psi = S_{\text{mod}}$$

design layer | **model layer: clusters i.i.d. within strata**

S_{des}
leading design term

Fixed-function design CLT, conditional on the finite population; consistency of the variance estimator

Lemma 1

T_1
empirical-process term

Controlled by complement stability, proved from the cluster model; needs only $r_m \to 0$

Lemma 4

T_2
second-order term

Neyman orthogonality: bounded by the product of the nuisance errors

Lemma 3

S_{mod}
census versus model

Negligible if $m/M_{\text{pop}} \to 0$; needed only for the super-population target Ψ

Lemma 2